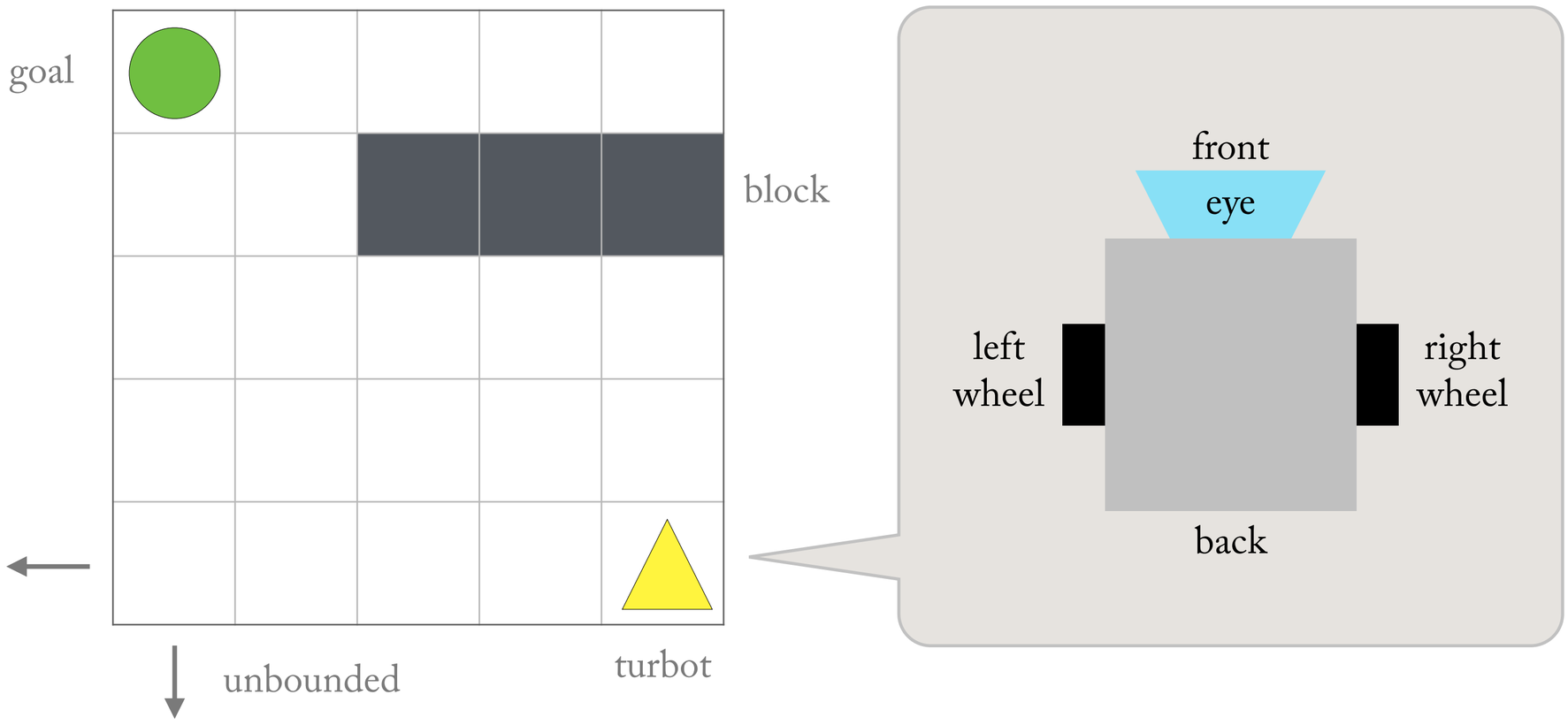
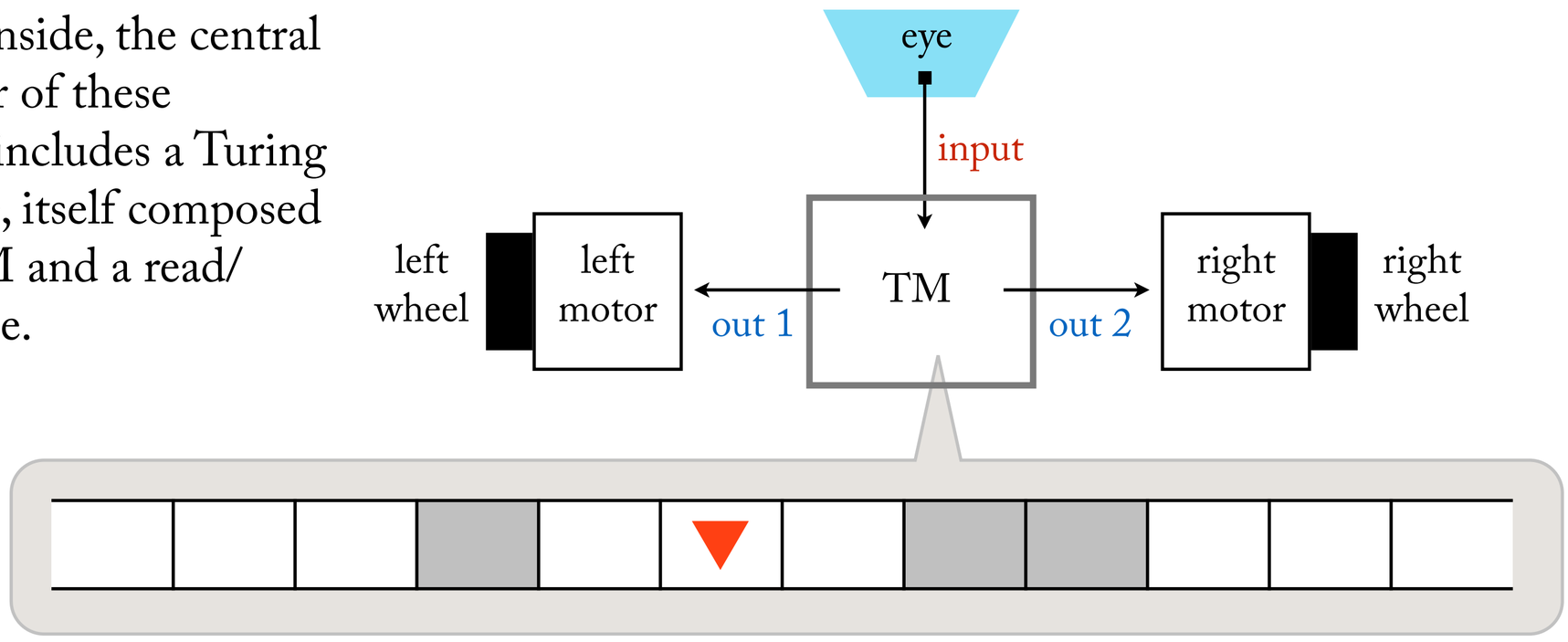
33. Computational Navigation

Turbots as Turing Machines

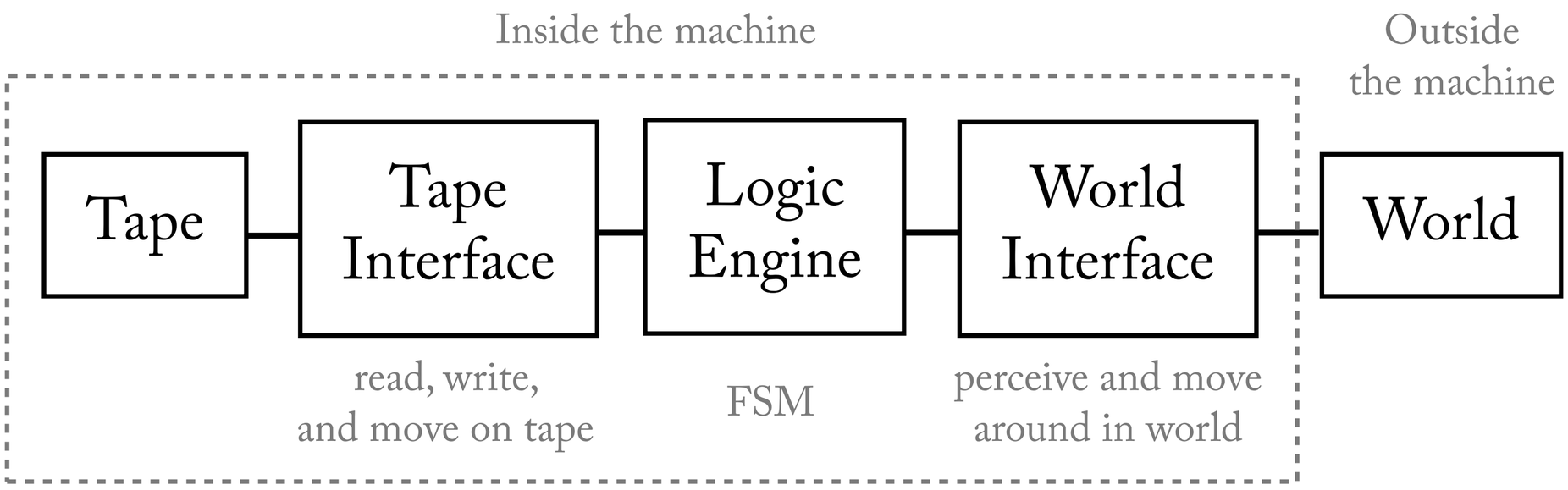
We next introduce Turbots based on Turing Machines. Viewed from the outside, they are just like the other navigational agents we've designed.



On the inside, the central processor of these Turbots includes a Turing Machine, itself composed of a FSM and a read/ write tape.



Turbots negotiate between two worlds: the *internal* world of the tape, and the *external* world of the environment. In this way they are just like you or me.



Turbots: Operation

- Turbots have an *internal* tape, but also act on an *external* world.

- At any moment, a Turbot has both an **internal position** (on the tape) and an **external position** (in the world).

- At each transition, a Turbot can perform either an internal operation, or an external operation, but not both.

- When a Turbot is performing an internal operation, its external position remains unchanged; when it is performing an external operation, its internal position remains unchanged.

- Turbots start with a blank tape (unless otherwise noted).

- Here are the operations available to Turbots:

Internal:



Input	**Output**
0 = read 0	0 = write 0
1 = read 1	1 = write 1
* = read *	* = write *
	R = move right
	L = move left

External:



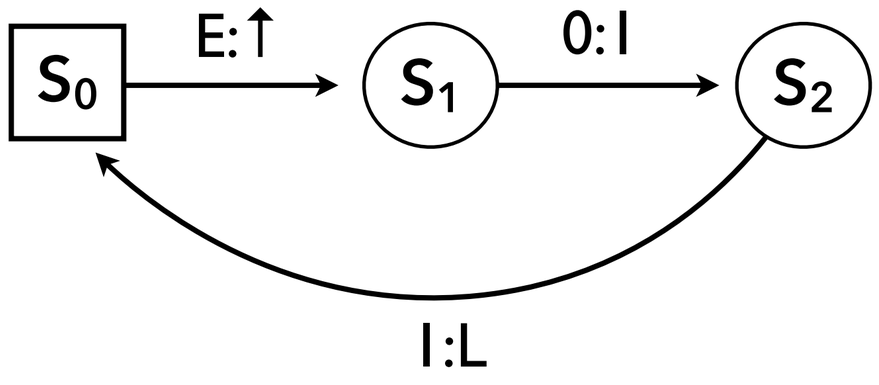
Input	**Output**
B = see block	↑ = move forward
E = see empty	↱ = right turn
F = see food	↰ = left turn

- By convention, states are either internal or external, but not both. External states are **square**, internal states are **circular**.

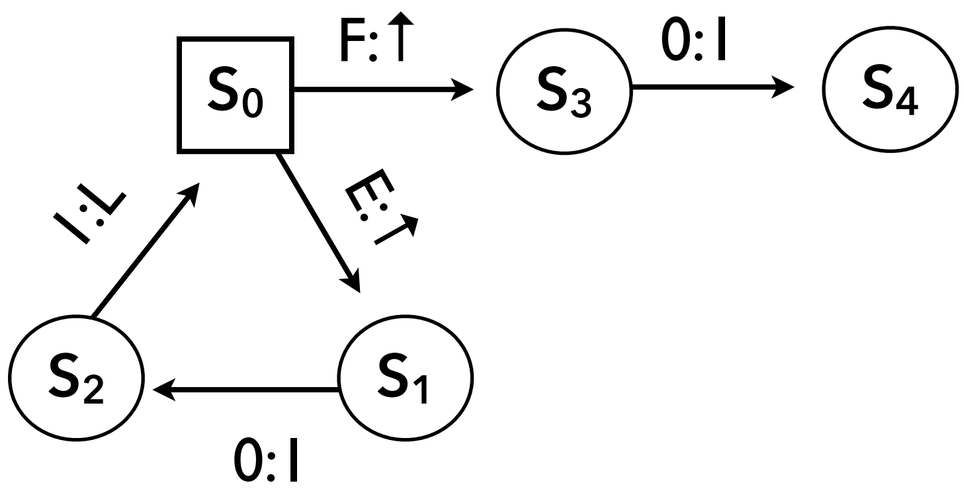
- The Turbot **cannot** pass through blocks. It **can** pass through food.

- Internally, a Turing Machine only perceives the cell it is **currently reading**.
- Externally, a Turbot only perceives the cell **in front of it**.

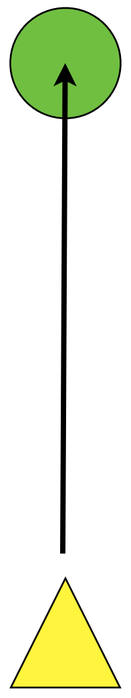
- This turbot moves forward, and records a "1" each time it does so.



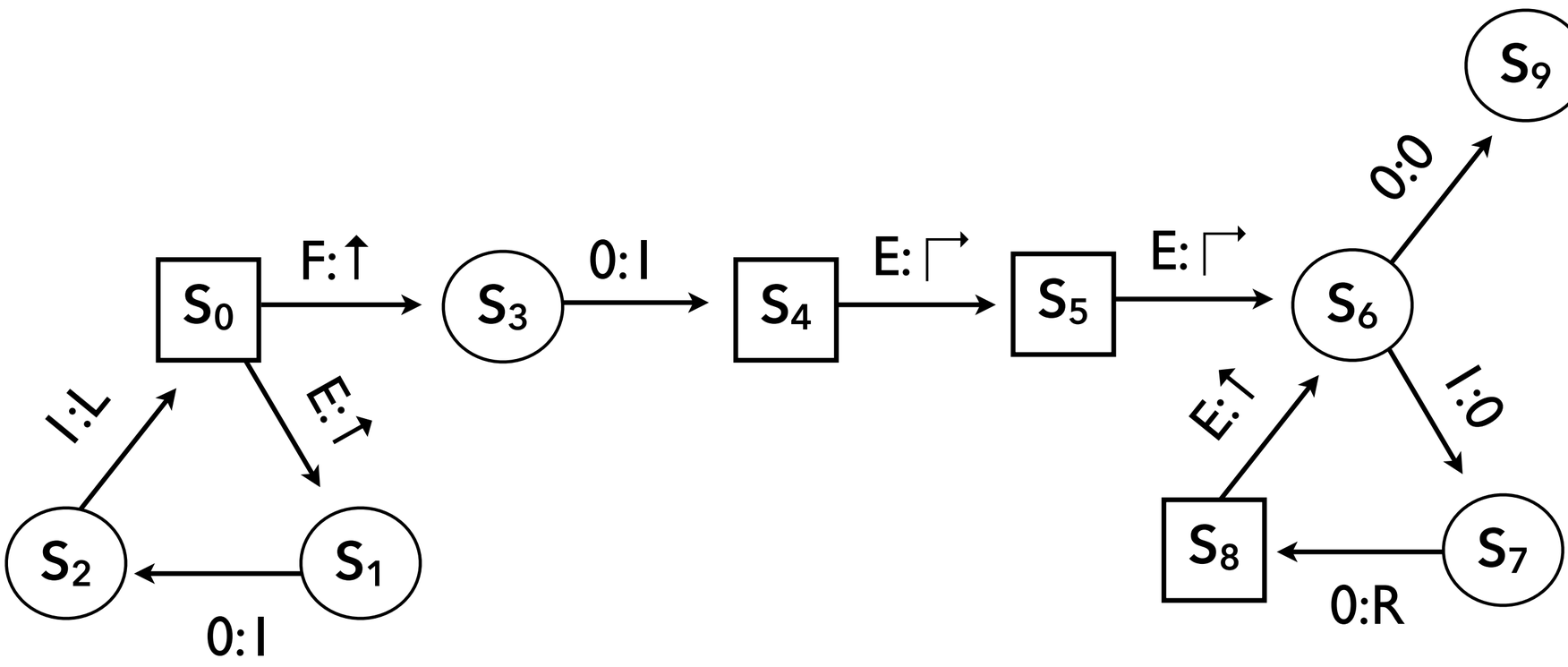
Half-Max Machine



Move to food directly in front of you, recording a "1" each time you move forward, then halt.



Mad-Max Machine



Move to food directly in front of you, but an unknown distance away, then return to the starting point and halt.

